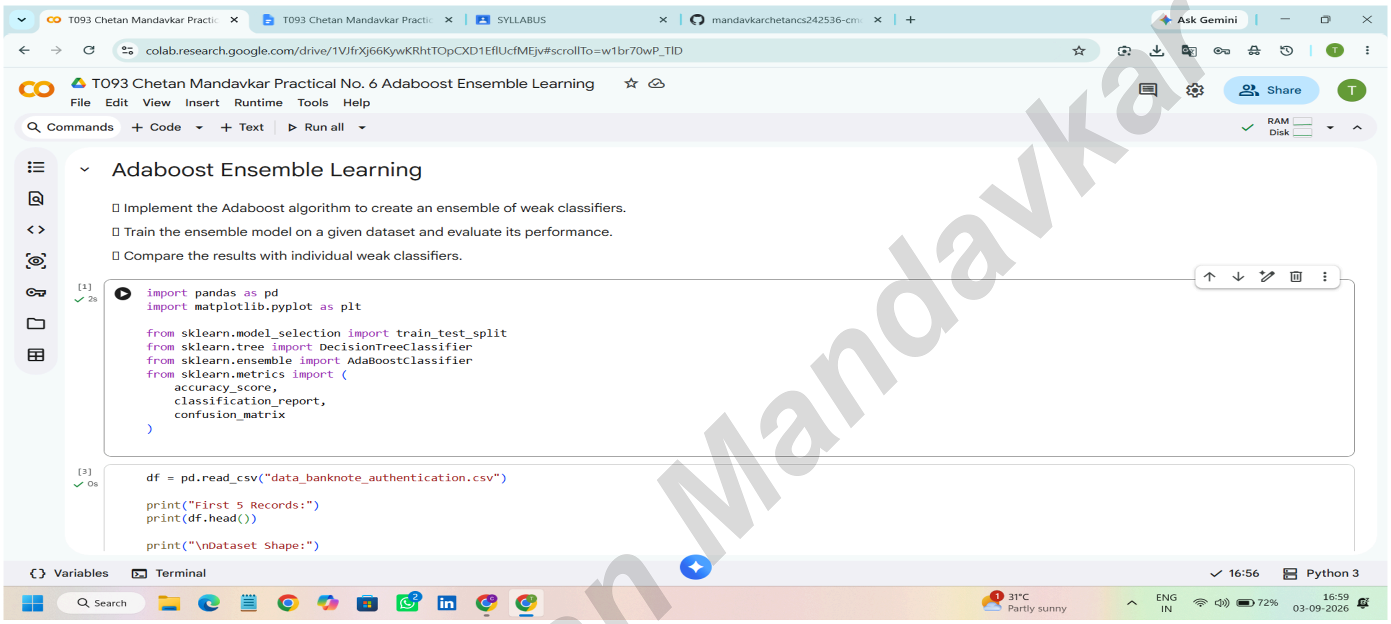
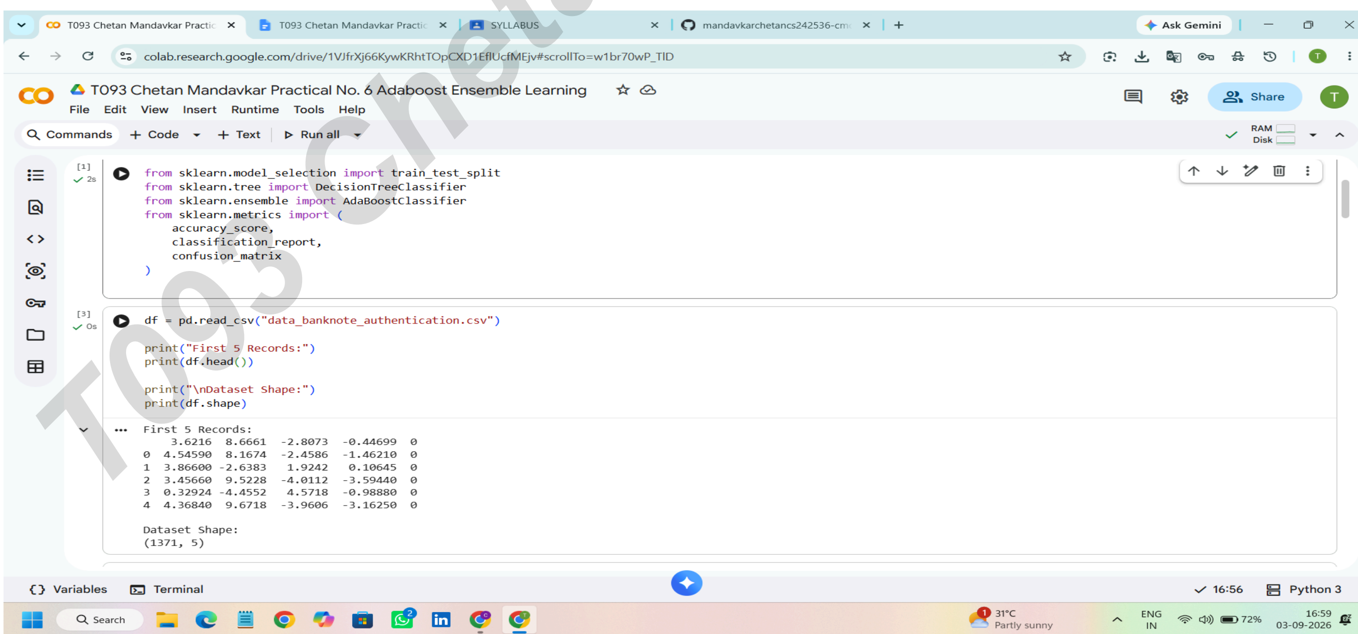
Sheth L.U.J. And Sir M.V. College

Aim:- Adaboost Ensemble Learning

Implement the Adaboost algorithm to create an ensemble of weak classifiers.Train the ensemble model on a given dataset and evaluate its performance.Compare the results with individual weak classifiers.



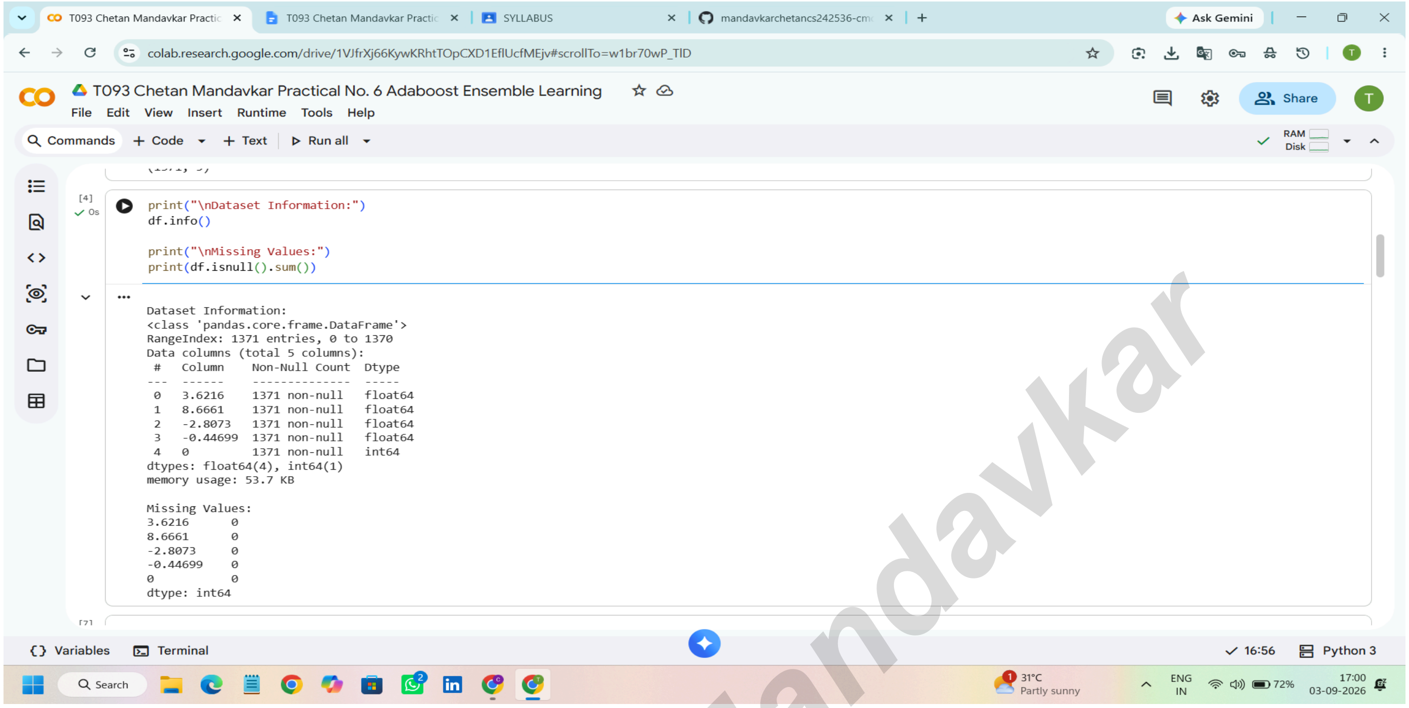


Name:- Chetan Mandavkar

Roll No. T093

Subject:- Artificial Intelligence

Sheth L.U.J. And Sir M.V. College



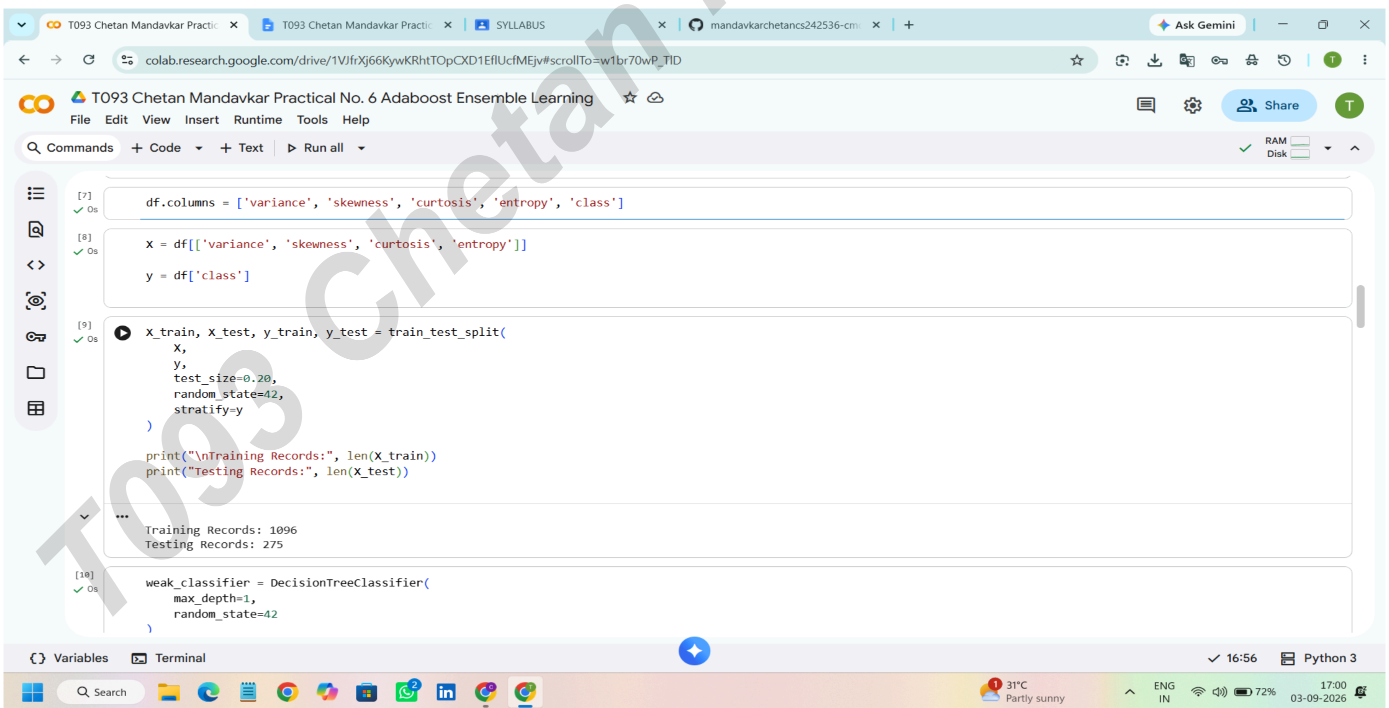
T093 Chetan Mandavkar Practical No. 6 Adaboost Ensemble Learning

```
[4]
print("\nDataset Information:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1371 entries, 0 to 1370
Data columns (total 5 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   3.6216    1371 non-null   float64
 1   8.6661    1371 non-null   float64
 2   -2.8073   1371 non-null   float64
 3   -0.44699  1371 non-null   float64
 4   0         1371 non-null   int64
dtypes: float64(4), int64(1)
memory usage: 53.7 KB

Missing Values:
3.6216      0
8.6661      0
-2.8073     0
-0.44699    0
0           0
dtype: int64
```



```
[7]
df.columns = ['variance', 'skewness', 'curtosis', 'entropy', 'class']

[8]
X = df[['variance', 'skewness', 'curtosis', 'entropy']]

y = df['class']

[9]
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("\nTraining Records:", len(X_train))
print("Testing Records:", len(X_test))
```

```
Training Records: 1096
Testing Records: 275
```

```
[10]
weak_classifier = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)
```


Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence

Sheth L.U.J. And Sir M.V. College

T093 Chetan Mandavkar Practical No. 6 Adaboost Ensemble Learning

[10]

```python
weak_classifier = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)


weak_classifier.fit(X_train, y_train)

weak_pred = weak_classifier.predict(X_test)


weak_accuracy = accuracy_score(
    y_test,
    weak_pred
)

print("\n----------------------------------------")
print("INDIVIDUAL WEAK CLASSIFIER")
print("----------------------------------------")

print("Accuracy:",
      weak_accuracy * 100, "%")
```

```
----------------------------------------
INDIVIDUAL WEAK CLASSIFIER
----------------------------------------
Accuracy: 85.81818181818181 %
```

T093 Chetan Mandavkar Practical No. 6 Adaboost Ensemble Learning

[11]

```python
ada_model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(
        max_depth=1,
        random_state=42
    ),
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)
```

[12]

```python
ada_model.fit(X_train, y_train)

print("\nAdaBoost Training Completed!")
```

```
AdaBoost Training Completed!
```

[13]

```python
ada_pred = ada_model.predict(X_test)
```

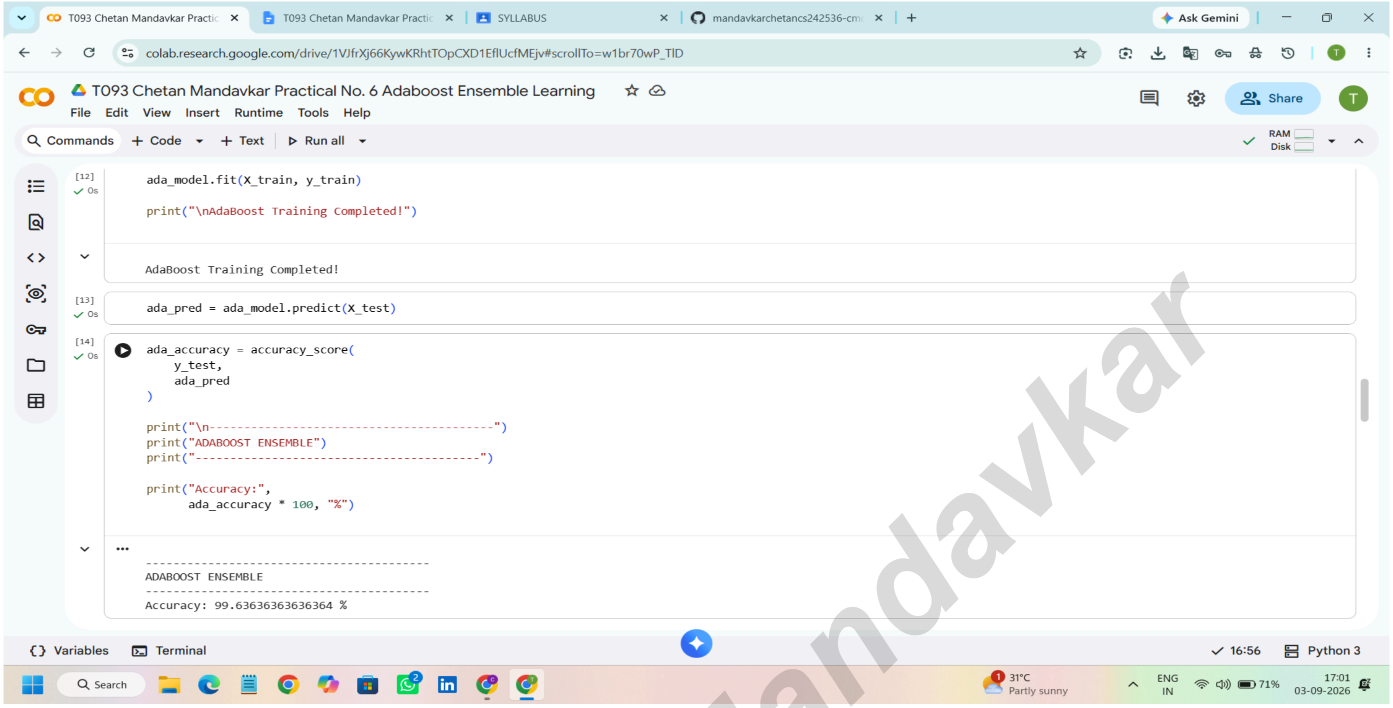
[14]

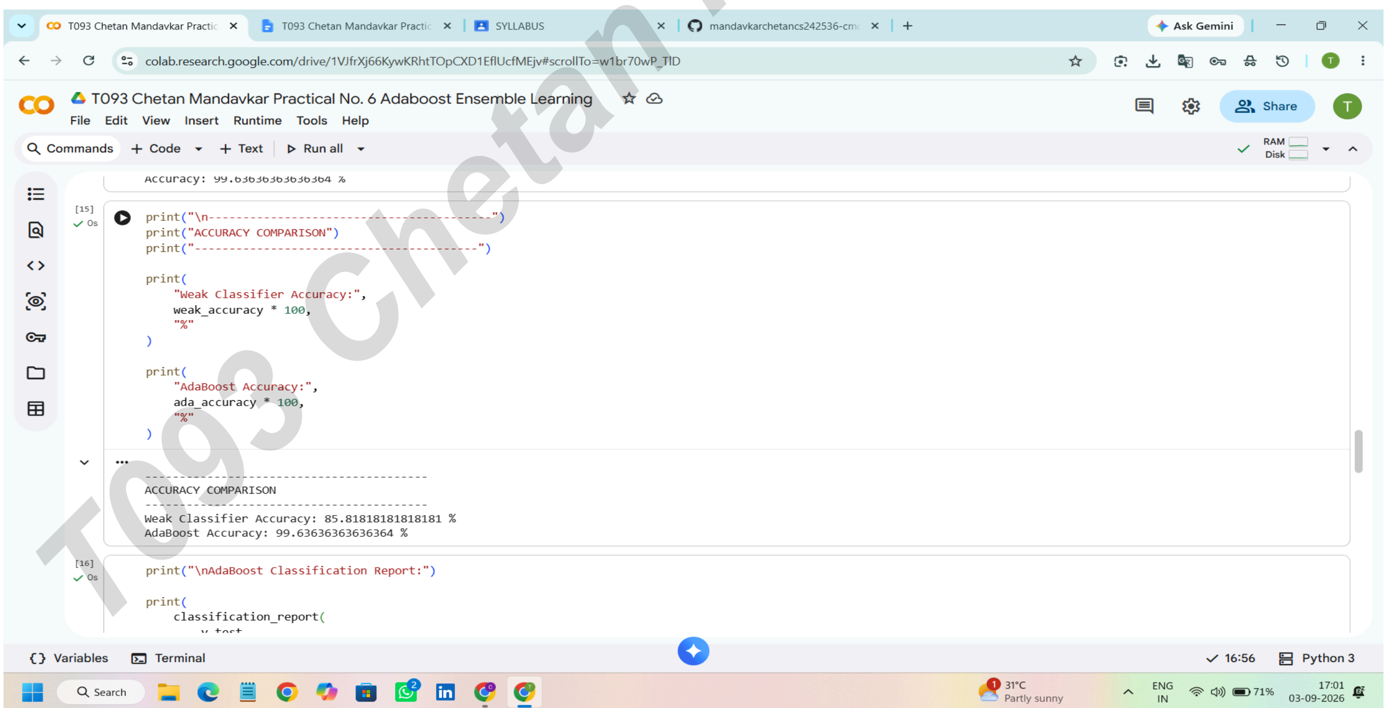
```python
ada_accuracy = accuracy_score(
    y_test,
    ada_pred
)

print("\n----------------------------------------")
print("ADABOOST ENSEMBLE")
```


Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence

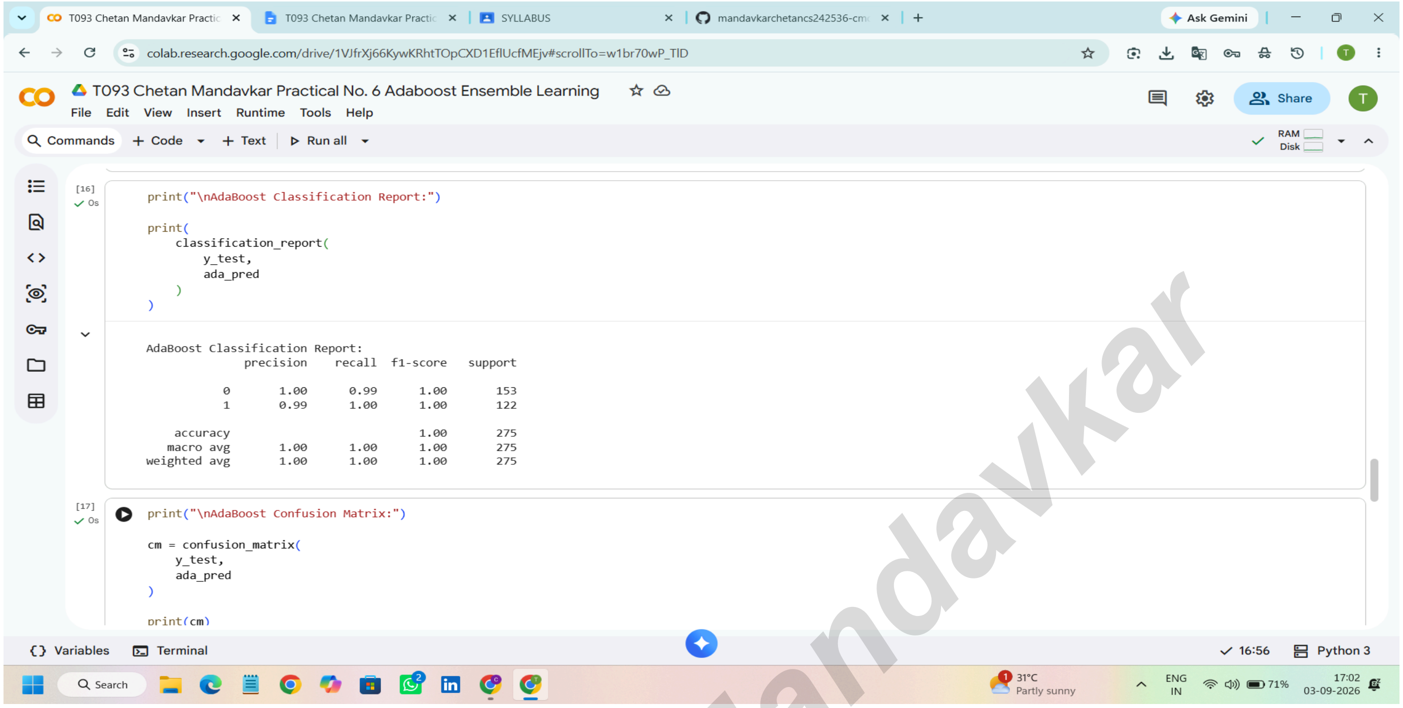
Sheth L.U.J. And Sir M.V. College

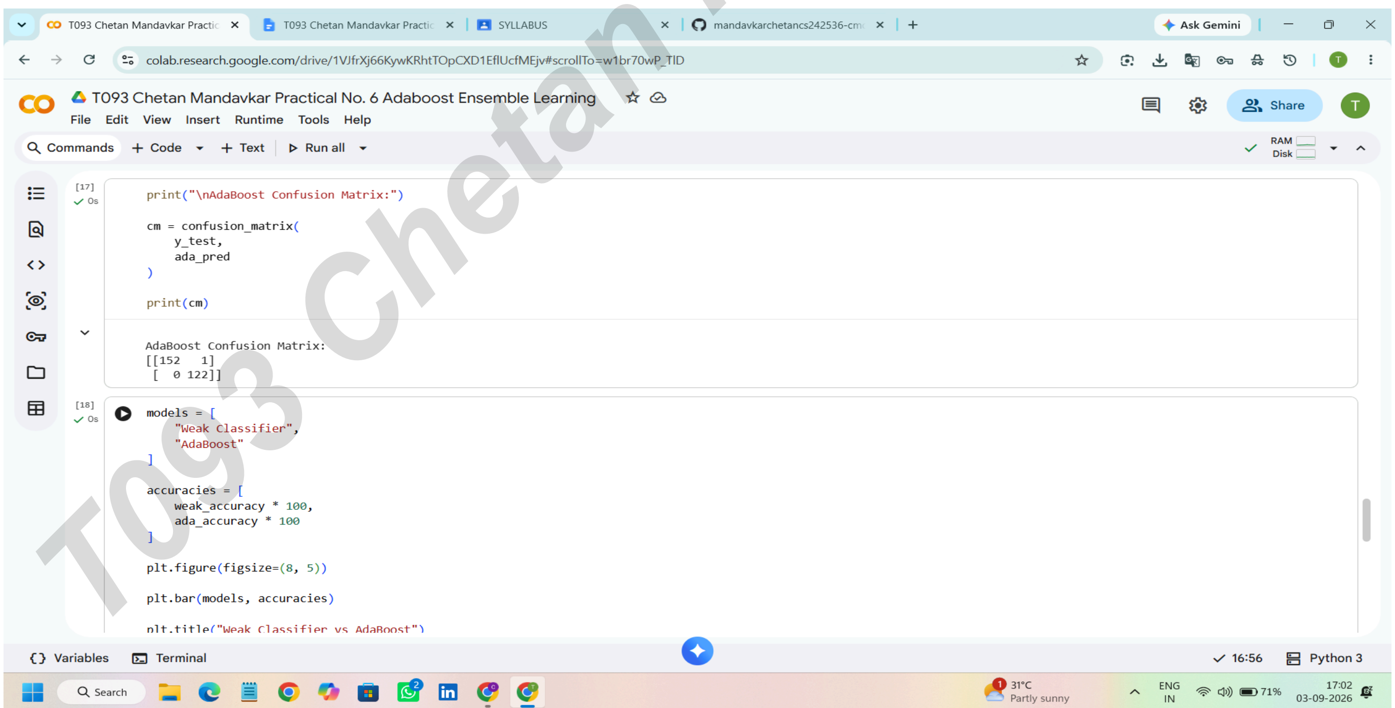




Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence

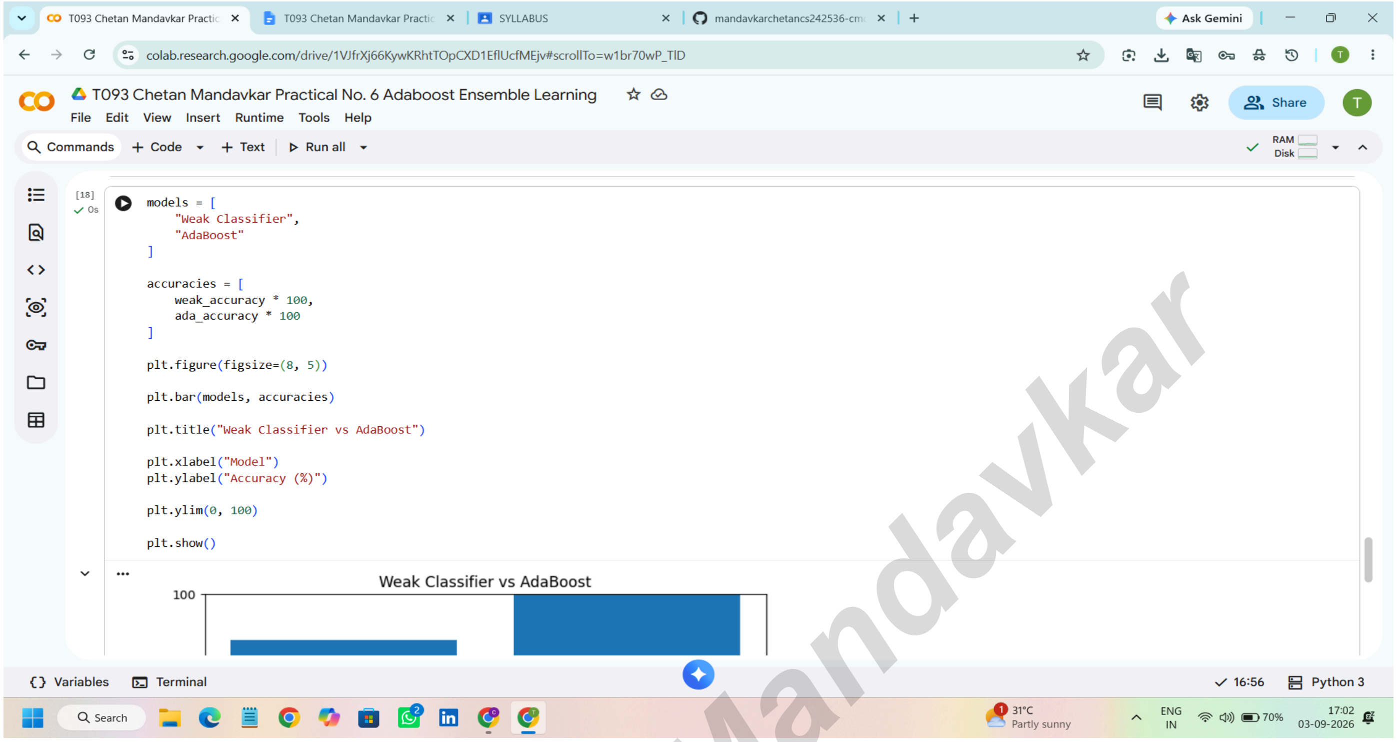
Sheth L.U.J. And Sir M.V. College

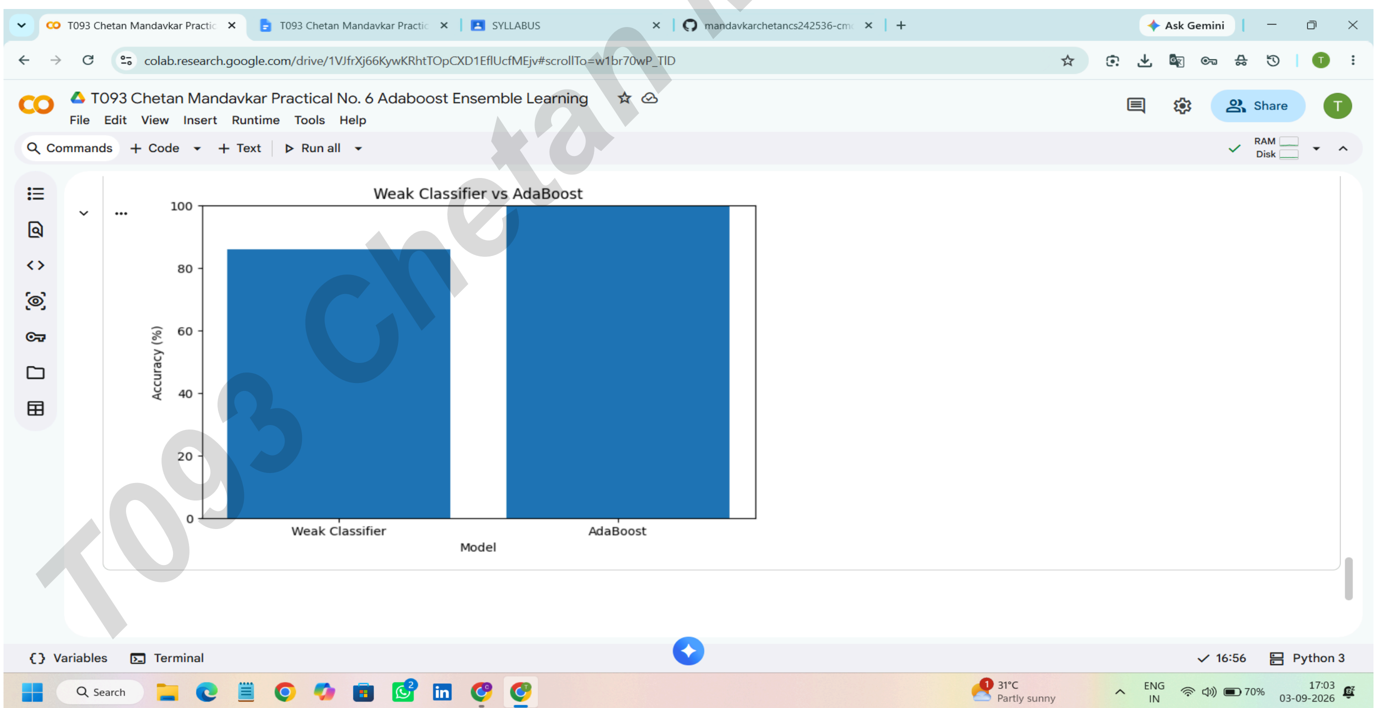




Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence

Sheth L.U.J. And Sir M.V. College





Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence